H-B-190
March 13, 1963
SUPERSEDING
(See 6.5)

FEDERAL SPECIFICATION

# BRUSHES, DUSTING

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers dusting brushes, with bristles set in wood or twisted wires.

**1.2 Classification.**

**1.2.1** ***Types, styles, classes, and sizes.*** Dusting brushes shall be of the following types, styles, classes and sizes, as specified (see 6.2 and 6.6).

Type I.—Brush, dusting, bench.

Style 1.—Bakery bench.

Style 2.—Counter (general use).

Class 1.—Hog bristle.
Class 2.—Horsetail hair.

Type II.—Brush, dusting, ceiling and wall.

Class 1.—Hog bristle.

Class 2.—Horsetail hair.

Size 1.—8 inch block.
Size 2.—12 inch block.

Type III.—Brush, dusting, draftsman's.

Type IV.—Brush, dusting, radiator.

Class 1.—Hog bristle, twisted-wire body, wood handle.

Class 2.—Horsetail hair, wood block and handle.

Class 3.—Tampico fiber, twisted-wire body, wood handle.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

*Federal Specification*:

PPP-B-636—Box, Fiberboard.

*Federal Standards*:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Hand-



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

books and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Standards*:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 **Standard product.** Brushes covered by this specification shall be standard products of the manufacturer except for such deviations from the manufacturer's standards as may be required by this specification.

3.2 **Figures.** Figures incorporated in this specification are included for convenience of identification and are not intended to preclude the purchase of brushes otherwise in accordance with the requirements of this specification.

3.3 **Preproduction sample.** When specified (6.2), before production is commenced, a sample of the finished item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.4 **Materials.** Material shall be as hereinafter specified. Materials not specifically defined shall be as regularly used by the manufacturer.

3.4.1 *Wood.* Wood for blocks and handles shall be close-grained and straight-grained hardwood such as beech, birch, or maple.

3.4.2 *Wire.*

3.4.2.1 *Twisting wire.* Twisting wire shall be not less than 0.109-inch-diameter wire, commercially identified as "brush handle wire"; hot-dip zinc-coated, electro-deposited zinc plated, or cadmium plated.

3.4.2.2 *Drawing wire.* Drawing wire shall be not less than 0.015-inch-diameter brass wire.

3.4.2.3 *Staple wire.* Staple wire shall be not less than 0.034-inch-diameter, commercial, steel-staple wire, coated to resist corrosion.

3.4.3 *Setting material.* Brush-filler setting material shall be high-quality brush maker's pitch or other suitable setting material.

3.4.4 *Brush filler.* Brush-filler material shall be bristle, fiber, or hair as hereinafter specified. Material shall be straight and resilient with no artificial coloring matter added. When tested as specified in 4.3.1, there shall be no perceptible change in stiffness or evidence of artificial color.

3.4.4.1 *Bristle.* Bristle shall be natural hog bristle, cleaned and sterilized as specified in 3.8.

3.4.4.2 *Fiber.* Fiber shall be tampico fiber, medium grade, natural and unbleached.

3.4.4.3 *Hair.* Hair shall be extra stiff, natural horsetail hair, cleaned and sterilized as specified in 3.8.

3.5 **Construction.** Construction shall be as hereinafter specified. The brush-filler material used to form the brush part shall conform to the requirements specified in 3.4.4



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

through 3.4.4.3, as applicable for the class brush required.

3.5.1 *Blocks.* Blocks shall be made from wood specified in 3.4.1 and shall be as hereinafter specified.

3.5.1.1 *Handle part.* The handle part of brushes shall be shaped to provide a comfortable handgrip. Unless otherwise specified herein, the handle end shall be provided with a hole not less than 3/16-inch diameter or other equally satisfactory means for hanging the brush by the handle when not in use.

3.5.1.2 *Brush part.* The brush part of blocks shall conform to the dimensional requirements of the applicable table. The drilled holes for brush-filler material shall be located so as to insure an even distribution of brush-filler material.

3.5.2 *Set.* Brush-filler material shall be set in the brush part in such a manner that, when tested as specified in 4.3.2, it shall not be possible to easily remove the brush-filler material from the brush part of staple-set, and twisted-in wire brushes. The pitch set and wire drawn brush-filler material shall not be removable without breaking the brush-filler material.

3.5.2.1 *Staple set.* The brush-filler material shall be cut to double length and center folded in quantities to insure individual tufts of such size to fit snugly, when inserted in the holes provided in the block. Stapling shall be accomplished so that the staples made from wire specified in 3.4.2.3 and brush-filler material are driven to the bottom of the holes, with the staple securely imbedded in the wood.

3.5.2.2 *Pitch set.* The brush-filler material shall be cut to suitable lengths and bound into tufts by not less than five turns of thread. The thread-bound end shall be dipped in setting material specified in 3.4.3 and redipped after knotting the thread. Tufts shall be of such size to fill the holes provided in the block and shall be secured in the holes with sufficient setting material to insure firmly set brush filler material. Not more than 1/8 inch of setting material shall be visible on the exposed brush-filler material of the finished brush.

3.5.2.3 *Twisted in wire.* The brush-filler material shall be cut to suitable length and securely held at the center in the twists of one twisting wire specified in 3.4.2.1, bent double. The brush-filler material shall be evenly distributed and shall form a spirally wound round brush head. The wire, at the brush end, shall form a loop not less than 1/4-inch diameter, for hanging.

3.5.2.4 *Wire, drawn.* Continuous length of drawing wire specified in 3.4.2.2 shall be looped up from the underside of the block in predrilled holes. Brush filler, cut to suitable length and of sufficient quantity to fill the holes provided, shall be secured at the centers in the drawn loops of wire. The drawing wire in the finished brush shall be concealed within the brush head.

3.5.3 *Trim.* The brush-filler material lengths shall be evenly and uniformly trimmed to conform to the dimensional requirements of the finished brushes as specified in tables I through V, as applicable.

3.5.4 *Weight.* Brush-filler material weight shall be as shown in tables I through V, as applicable, when tested as specified in 4.3.3.

3.6 **Finish.** Wood handles and blocks, except type IV, shall be coated with varnish, lacquer, or other suitable clear organic material in accordance with the manufacturer's regular commercial practice. At the option of the manufacturer, type IV handles may be furnished with a plain, smooth sanded finish.

3.7 **Marking for identification.** Unless otherwise specified (6.2), in addition to any special marking required by the contract or order, the contractor shall permanently indent his name or trademark of such known



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

character as to be easily identifiable with said manufacturer on the handle of each brush. The indented marking shall be coated with a commercial, waterproof stencil paint.

3.8 **Sterilization.** Horsetail hair and hog bristle shall be sterilized after thorough washing and rinsing, in accordance with any one of the procedures specified in 3.8.1, 3.8.2, or 3.8.3. A properly checked steam-pressure gauge (applicable for the method in 3.8.3) and a thermometer (for all methods), both visible from the outside of the chambers, shall be provided with a suitable recording device to provide a record of the pressure, when applicable, and temperature of the disinfecting chamber.

3.8.1 *Boiling water.* Hog bristle and horsetail hair shall be soaked in boiling water (212°F.) for not less than 30 minutes.

3.8.2 *Dry heat.* Hog bristle and horsetail hair shall be placed in a chamber and a temperature above 284°F. shall be developed and maintained for not less than 3 hours.

3.8.3 *Autoclaving.* Horsetail hair and hog bristle shall be placed in a pressure chamber for a minimum of 20 minutes at not less than 15 pounds pressure per square inch at a temperature not less than 248°F.

3.9 Detailed requirements.

3.9.1 *Type I brushes.* Type I brushes shall conform to the applicable requirements of table I for the style and class brush required. Design shall be similar to figures 1 or 2 as applicable. The block and handle shall be one integral piece, having a well-rounded back along the brush part of the block.

3.9.1.1 *Type I, style 1, bakery-bench brush.* Bakery-bench brush-filler material shall be staple set in accordance with 3.5.2.1 The end of the brush part of the block shall be beveled upward to insure that the brush-filler material extends beyond the brush not less than one inch.

3.9.1.2 *Type I, style 2, counter brush (general use).* Class 1 brush-filler material shall be pitch set in accordance with 3.5.2.2. At the option of the manufacturer, class 2 brush-filler material shall be pitch set in accordance with 3.5.2.2, or set with staples in accordance with 3.5.2.1.

3.9.2 *Type II, ceiling and wall brush.* Type II brushes shall conform to the applicable requirements of table II for the size brush required. Design shall be similar to figure 3. Brush-filler material shall be staple set in accordance with 3.5.2.1, or pitch set in accordance with 3.5.2.2.

3.9.2.1 *Block.* The wood block for sizes 1 and 2 brushes shall be 2 inches, plus or minus ⅛ inch, in diameter at the center and tapering in diameter to 1-½ inches, plus or minus ⅛ inch, in diameter at the ends. Overall length shall be as shown in table II.

3.9.2.2 *Holes.* The holes in the brush part of the block for type II brushes shall be placed at an angle to produce an even distribution of bristle or hair throughout the brush, and a lateral spread of bristle or hair of 5-1/4 inches, plus or minus 1/4 inch, in the portion between the ball ends. The holes shall be located entirely around each end of the block, covering a distance of approximately 1-3/8 inches from the ends, in such a manner as to produce a brush with rounded or ball ends. In the center of the block, there shall be a 3/4-5 Acme thread tapped handle hole.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE I.- Type I, styles 1 and 2, bakery bench and counter brush (general use), requirements

| Requirement | Unit of measure | Style 1 | Style 2 | | |
|---|---|---|---|---|---|
| | | Classes 1 and 2 staple set | Class 1 pitch set | Class 2 staple set | Class 2 pitch set |
| Block: | | | | | |
| Length, overall (min.) | inches | 13 | 13 | 13 | 13 |
| Length, brush part (min.) | inches | 8 | 8 | 8 | 8 |
| Width, brush part | inches | 1-3/4 ± 1/16 | 1-3/4 ± 1/16 | 1-3/4 ±1/16 | 1-3/4±1/16 |
| Thickness, brush part | inches | 1-1/16 ± 1/16 | 1-1/16 ± 1/16 | 1-1/16 ± 1/16 | 1-1/16±1/16 |
| Holes: | | | | | |
| Number: | | | | | |
| In outside rows (min) | | 41 | 34 | | 34 |
| In inside rows (min) | | 47 | 24 | | 24 |
| Total (min) | | 88 | 58 | 85 | 58 |
| Number of rows | | | | 5 | |
| Diameter: | | | | | |
| Inside rows (min) | inch | 3/16 | 5/16 | 13/64 | 1/4 |
| Outside rows (min) | inch | 3/16 | 3/16 | 13/64 | 3/16 |
| Depth (min) | inch | 5/16 | 15/32 | 5/16 | 15/32 |
| Bristle: | | | | | |
| Length clear of block (min) | inches | 3 | 2-3/4 | | |
| Weight (min) | ounces | 3 | 3 | | |
| Spread of bristles (approx) | inches | 4 | 4-1/2 | | |
| Hair: | | | | | |
| Length clear of block (min) | inches | 3-1/4 | | 2-3/8 | 2-3/4 |
| Weight (min) | ounces | 4 | | 4 | 4 |
| Spread of hair (approx) | inches | 4 | | 4 | 4 |

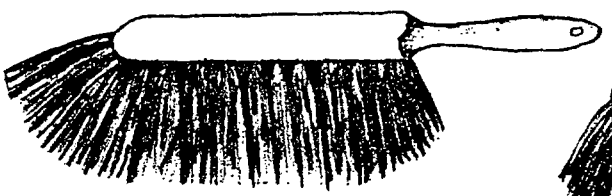


FIGURE 1.

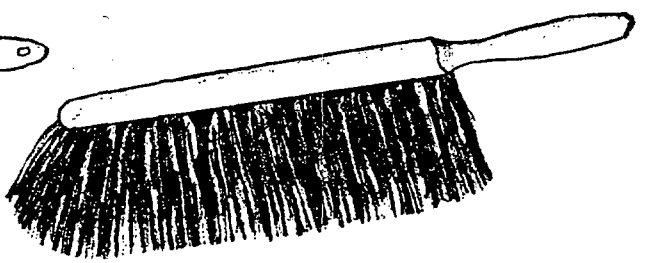


FIGURE 2.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

3.9.2.3 *Handle.* When specified (6.2), handles for type II brushes shall be required. Handle length shall be as specified (6.2). A tolerance of plus or minus one inch in handle length will be permitted. The handle shall be one-piece or two-piece joint construction. One end shall have a 3/4-5 Acme thread for a distance of not less than 1-1/16 inches to fit the tapped hole in the block.

3.9.3 *Type III, draftsman's brush.* Type III brushes shall conform to the requirements of table III. Design shall be similar to figure 4. Brush-filler material shall be wire drawn or staple set in accordance with 3.5.2.1 or 3.5.2.4, as applicable. Handle and block shall be one piece, wood construction.

TABLE II.- Type II, ceiling and wall brush, requirements

| Requirement | Unit of measure | Size 1 | Size 2 |
|---|---|---|---|
| Block: | | | |
| Length | inches | 8 ± 1/4 | 12 ± 1/4 |
| Holes for bristle or horsehair: | | | |
| Depth, minimum | inch | 3/8 | 3/8 |
| Diameter of holes, outside rows, minimum | inch | 7/32 | 7/32 |
| Diameter of holes, inside rows, minimum | inch | 9/32 | 9/32 |
| Number of holes, outside rows, minimum | | 42 | 54 |
| Number of holes, inside rows, minimum | | 70 | 110 |
| Hole for handle: | | | |
| Depth | inches | 1-1/2 | 1-1/2 |
| Handle: | | | |
| Diameter | inch | 7/8 ± 1/32 | 7/8 ± 1/32 |
| Bristle: | | | |
| Length clear of block | inches | 2-7/8 ± 1/16 | 2-7/8 ± 1/16 |
| Weight of bristle per brush, minimum | ounces | 6.0 | 9.0 |
| Horsehair: | | | |
| Length clear of block | inches | 2-3/8 ± 1/16 | 2-3/8 ± 1/16 |
| Weight of horsehair per brush, minimum | ounce | 6-1/4 | 9-1/3 |

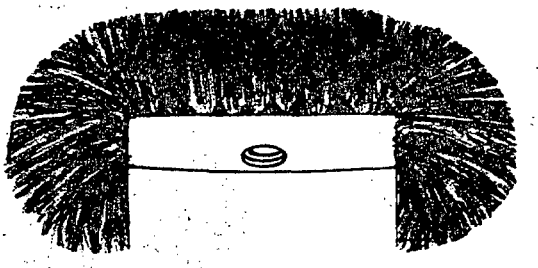


FIGURE 3.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE III.- Type III, draftsmen's brush, requirements

| Requirement | Unit of measure | Measure |
|---|---|---|
| Block: | | |
| Length, over-all | inches | 13-1/2 ± 1/4 |
| Length, brush part | inches | 8 ± 1/8 |
| Width, brush part | inches | 1-1/4 + 0 -1/2 |
| Thickness, brush part | inches | 5/8 + 0 - 1/4 |
| Holes: | | |
| Number, minimum | | 25 (1 row) |
| Diameter, minimum | inch | 3/16 |
| Depth, minimum | inch | 5/16 |
| Spacing, center to center, approx. | inch | 5/16 |
| Location | | Middle of brush part of block |
| Hair: | | |
| Weight, minimum | ounces | 1.0 |
| Length, clear of block | inches. | 2-1/4 ± 1/4 |

TABLE IV.- Type IV, classes 1 and 3, radiator brush, requirements

| Requirement | Unit of measure | Measure |
|---|---|---|
| Length, over-all | inches | 20-1/2 ± 1/2 |
| Brush head: | | |
| Diameter | inches | 2-3/8 ± 1/16 |
| Length | inches | 7 ± 1/4 |
| Handle: | | |
| Diameter | inches | 1 ± 1/16 |
| Length | inches | 4-1/2 |
| Bristle or Fiber: Weight, minimum | ounces | 7/16 |

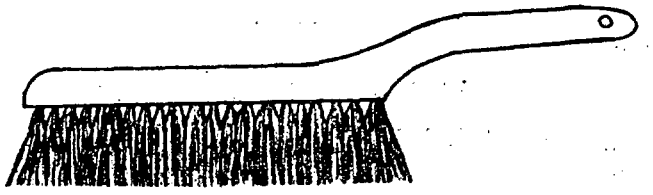


FIGURE 4.

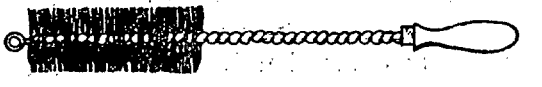


FIGURE 5.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE V .- Type IV, class 2, radiator brush, requirements

| Requirement | Unit of measure | Measure |
|---|---|---|
| Block: | | |
| Length, over-all | inches | 24 or 32 ± 1/4 |
| Width, center of brush part | inches | 1-1/8 ± 1/8 |
| Thickness, brush part | inch | 3/8 ± 1/16 |
| Holes: | | |
| Diameter, minimum | inch | 11/64 |
| Depth, minimum | inch | 5/16 |
| Number, minimum | | 32 |
| Number of rows | | 1 |
| Hair: | | |
| Weight, minimum | ounces | 1.21 |
| Length clear of block, minimum | inches | 2-1/8 |



FIGURE 6.

3.9.4 *Type IV, radiator brush.*

3.9.4.1 *Type IV, classes* 1 *and* 3 *twisted in wire brush.* Twisted in wire brushes shall conform to the requirements of table IV. Design shall be similar to figure 5.

3.9.4.1.1 *Handle.* Handle for type IV, classes 1 and 3 brushes shall be wood securely attached to the extension of the twisted wires in such a manner that the handles will not pull free of the wires, when tested as specified in 4.3.4. The end of the handle shall be fitted with a screw eye, or shall have a through hole not less than 3/16-inch diameter located 3/4 inch, plus or minus 1/4 inch, from the free end. As an alternate construction, one end of the body wire shall extend through the handle and shall be turned back to form an eye for hanging.

3.9.4.2 *Type IV, class* 2, *wood block and handle.* Type IV, class 2 brushes shall conform to the requirements of table V. Design shall be similar to figure 6. Handle and block shall be one-piece wood construction. Brush material shall be wire drawn or staple set in accordance with 3.5.2.1 or 3.5.2.4, as applicable.

3.10 **Workmanship.** The brushes shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order (6.2). The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling and inspection for acceptance.** Except when otherwise specified by Civil Government agencies (6.2) sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. For purpose of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type, style, class, and size offered for delivery at one time.

4.2.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.1.1 *Certification of compliance.* Components and materials listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in applicable paragraphs of this specification.

| *Component* | *Characteristic* | *Requirement paragraph* |
|---|---|---|
| Horsetail hair | Sterilization | 3.8 |
| Hog bristle | Sterilization | 3.8 |

4.2.2 *Process examination.* Examination shall be made of the composition and thickness of drawing wire, the thickness and coating of staple wire, the depth of holes for brush-filler material, and the binding, dipping, and setting of pitch-set brushes, to establish that no deviation is made from the specified procedure or requirement. Whenever a deviation is noted, correction shall be made. Failure to make immediate correction shall be cause for rejection of affected end-item lots.

4.2.3 *Inspection of the end item.*

4.2.3.1 *Examination of the end item.* Examination of the end item shall be in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of brushes for examination under 4.2.3.1.1 and 4.2.3.1.2, in units of intermediate containers for examination under 4.2.4.1, and in units of shipping containers for examination under 4.2.4.2.

4.2.3.1.1 *Examination of the end item for defects in material, construction and workmanship.* The sample unit for this examination shall be one brush.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

| Examine | Defect |
|---|---|
| Materials: Wood (blocks and handles) | Not a close-grained and straight-grained hardwood. |
| Twisting wire | Not zinc or cadmium plated. |
| Bristle | Not natural hog bristle; not stiff and resilient. |
| Fiber | Not natural, unbleached, tampico fiber. |
| Hair | Not extra-stiff horsetail hair. |
| Construction and appearance | Handle portion of brush not shaped to provide comfortable handgrip. |
| | Handle end not provided with hole or other means for hanging brush. |
| | Wood component not finished with varnish, lacquer or suitable clear organic material. |
| | Brush-filler material not securely held in block or twisted-wire handle; can be easily removed by hand. |
| | Exposed sharp wire, nails, sharp edges or splinters. |
| Type I brush | Not of the general design as shown on figures 1 or 2, as applicable. |
| | Not class specified. |
| | Block and handle not one integral part; back of brush part of block not well rounded. |
| Style I | End of brush part of block not beveled upward; brush-filler material does not extend at least one inch beyond end of block. |
| Type II brush | Not of the general design as shown on figure 3; does not have rounded or ball ends. |
| | Not class specified. |
| | Block does not have hole in center for attaching handle; hole not threaded with 3/4-5 Acme thread. |
| | Handle (when specified) not one-piece construction or two-piece jointed construction; handle not threaded on one end with Acme thread; does not snugly engage thread of block. |
| Type III brush | Not of the general design as shown on figure 4. |
| | Not class specified. |
| | Handle and block not one-piece construction. |
| Type IV brush | Not of the general design as shown on figures 5 or 6 as applicable. |
| Classes 1 and 3 | Not formed from one piece of twisting wire, bent double, as specified; brush-filler material does not form spirally wound, round brush head. |
| | Wood handle not securely held on extension of twisted wire. |
| Class 2 | Handle and block not one-piece construction. |
| Workmanship | Brush-filler material not evenly distributed in brush head. |
| | Brush-filler material not clean; evidence of dirt or foreign matter. |
| | Brush head not evenly trimmed. |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

| Examine | Defect |
|---|---|
| Class 2 Workmanship (cont'd) | Wood component not smoothly finished; splintered, torn or loose grain. |
| | Threads on block or handle stripped or damaged (type II only). |
| | Sharp edge on wire which might cause injury. |
| | Identification marking omitted, incomplete, incorrect or not as specified. |

4.2.3.1.2 *Dimensional examination.* Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension not within the specified requirements shall be classified as a defect. The sample unit for this examination shall be one brush.

| Examine | Defect |
|---|---|
| All types | Any dimension not within the tolerance specified for the applicable type, style, class or size (see tables I through V). |
| | Number of holes less than specified in applicable table. |
| Type II only | Center of wood block not 2 inches in diameter, tapering to 1-1/2 inches in diameter (a tolerance of plus or minus 1/8 inch is allowed). |
| | Lateral spread of bristle or hair in portion between the ball ends less than 5 inches or greater than 5-1/2 inches. |
| | Handle (when specified) varies from specified length by more than plus or minus 1 inch. |

4.2.3.1.3 *Inspection levels and acceptable quality levels (AQL's) for examination.* The inspection levels for determining the sample size and the acceptable quality levels (AQL's), expressed in defects per hundred units, shall be as follows:

| *Examination paragraph* | *Inspection level* | *AQL* |
|---|---|---|
| 4.2.3.1.1 | I | 4.0 |
| 4.2.3.1.2 | L 2 | 2.5 |
| 4.2.4.1 | L 4 | — |
| 4.2.4.2 | L 4 | 4.0 |

4.2.3.1.4 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table VI herein, from each lot presented for examination of each type, style, class, and size brush. The sample unit shall be one brush and the lot size shall be expressed in units of brushes. The inspection level and acceptable quality level shall be as indicated in table VI.

4.2.4 *Examination of preparation for delivery.*

4.2.4.1 *Examination of the end item for count.* The sample unit for this examination shall be one intermediate container. Not less than three sample units shall be selected for this examination. The average count per container shall be not less than the specified or indicated quantity.

4.2.4.2 *Examination of preparation for delivery (except count).* An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified below. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE VI.- Instructions for testing

| Characteristic | Specification reference | | Requirements applicable to | | No. determinations per unit | Results reported as | | Inspection level | AQL |
|---|---|---|---|---|---|---|---|---|---|
| | Requirement | Test method | Individual unit | Lot average | | Pass or fail | Numerically to nearest | | |
| Tightness of filler material | 3.5.2 | 4.3.2 | X | - | 10 | X | - | L 3 | 4.0 |
| Weight of bristle, fiber and hair | 3.5.4 and Tables I thru V | 4.3.3 | - | X | 1 | - | 0.1 ounce | L 3 | - |
| Soak test | 3.4.4 | 4.3.1 | X | - | 1 | X | - | L 3 | 4.0 |
| Pull test for handle | 3.9.4.1.1 | 4.3.4 | X | - | 1 | X | - | L 3 | 4.0 |

| Examine | Defect |
|---|---|
| Packaging | Not unit packaged as specified. |
| Packing | Any packing component or material missing, damaged or not as specified; closure not accomplished by specified or required method or materials. |
| Workmanship | Inadequate application of component or material such as incomplete closure of container flap, tape or case liner, bulge or distortion of container. |
| Content | Number of unit or intermediate packages not in accordance with requirements. |
| Marking | Omitted, incomplete, incorrect, illegible; not in accordance with contract requirements. |

4.3 Tests.

4.3.1 *Soak test.* Bristle, fiber, and hair shall be soaked in boiling water (212°F.) for not less than one hour and examined for compliance with 3.4.4.

4.3.2 *Brush-filler material tightness.* Individual strands of brush-filler material shall be held between the thumb and forefinger and pulled to determine compliance with 3.5.2.

4.3.3 *Bristle, fiber, and hair weight.* Bristle, fiber, or hair shall be removed from the brush and conditioned for not less than 16



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

hours in air at a relative humidity of 50, plus or minus 4 percent, and at a temperature of 23, plus or minus 1.1 degrees centigrade. Weighing for compliance with 3.5.4 shall be performed in the same atmosphere.

4.3.4 *Pull test for handle.* With the loop end of the brush securely attached to an immovable object, an attempt shall be made to pull the handle free of the wire by hand.

## 5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packaging. Packaging shall be level A, B, or C, as specified (6.2).

5.1.1 *Level A or B.* Twelve brushes of one type, style, class, and sizes only shall be packaged in a snug-fitting fiberboard box conforming to type I or II, class 1, style RSC of PPP-B-636. Type I and type II brushes shall be arranged in two layers of six each with pads of fiberboard between each layer. Type I brushes shall be alternated head to handle in each layer. All other brushes shall be arranged in the most compact manner. Corrugated-fiberboard layer pads shall be 1/2 inch less than the inside length and width dimensions of the fiberboard box. Not less than 4 grams of naphthalene in ball or flake form shall be placed in each box, except boxes containing type IV, class 3 brushes. All flaps of the fiberboard boxes shall be securely sealed with an adhesive commercially used for the specific product being packaged by application throughout the entire area of contact between the flaps.

5.1.2 *Level C.* Brushes shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Brushes of one type, style, class, and size only, packaged as specified in 5.1, shall be packed in the quantities specified in table VII in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed with a case liner or tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.*

5.2.2.1 *Brushes, types I and II.* Type I and type II brushes of one style, class, and size only, packaged as specified in 5.1.1, shall require no further packing.

Table VII.- Packing of brushes

| Brush | Quantity per shipping container |
|---|---|
| Type I | 36 |
| Type II | 24 |
| Type III | 96 |
| Type IV | 72 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

5.2.2.2 *Brushes, type III and IV.* Ninety-six type III brushes and 72 type IV brushes of one class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636, except that the minimum bursting strength of the fiberboard shall be 275 p.s.i. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed, by application throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Brushes of one type, style, class, and size only, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use. The brushes covered by this specification are intended to be used for dusting operations as indicated by the nomenclature of the various brushes covered.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, style, class, and size of brush required (1.2).

(*c*) When preproduction sample is required (3.3).

(*d*) Identification marking required if other than specified (3.7).

(*e*) When handles are required for type II, ceiling and wall brushes (3.9.2.3).

(*f*) Handle length required (3.9.2.3).

(*g*) Inspection record details (4.1).

(*h*) When sampling and inspection shall be performed in accordance with other than the provisions set forth in MIL-STD-105 (4.2).

(*i*) Applicable levels of packaging and packing (5.1 and 5.2).

6.3 Illustrations. The illustrations shown herein as figures 1 thru 6 are descriptive and not restrictive, and are not intended to preclude the purchase of brushes otherwise conforming to this specification.

6.4 Standard Pack (civil agencies). The Standard Pack for this commodity shall be as specified in 5.1.1 and 5.2.2. The Standard Pack requirements are intended for use in procurements of stores stock replenishments. Procuring officers shall use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or immediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

6.5 **Supersession data.** This specification supersedes the following specifications:

| *Number* | *Title* | *Date* |
|---|---|---|
| H-B-189 | Brushes, Dust; Bakery Bench | Mar. 14, 1947 |
| H-B-191a | Brushes, Dust; Ceiling and Wall | Oct. 8, 1947 |
| H-B-201b | Brushes, Dust; Counter (General use) | Nov. 6, 1947 |
| H-B-204 | Brushes, Dust; Draftsmen's | Feb. 27, 1947 |
| H-B-221a | Brushes, Dust; Radiator | Mar. 14, 1947 |

6.6 **Classification comparison.** Classification comparison between superseded specifications and this specification is as follows:

| *Superseded Specifications* | | *Specification H-B-190* |
|---|---|---|
| H-B-189 ... | Grade A | Type I, Style 1, Class 1 |
| H-B-189 ... | Grade B | Type I, Style 1, Class 2 |
| H-B-191a .. | Grade A, Size 1 | Type 2, Class 1, Size 1 |
| H-B-191a .. | Grade A, Size 2 | Type 2, Class 1, Size 2 |
| H-B-191a .. | Grade B, Size 1 | Type 2, Class 2, Size 1 |
| H-B-191a .. | Grade B, Size 2 | Type 2, Class 2, Size 2 |
| H-B-201b .. | Grade A | Type I, Style 2, Class 1 |
| H-B-201b .. | Grade B | Type I, Style 2, Class 2 |
| H-B-204 ... | — | Type III |
| H-B-221a, Type I .. | — | Type IV, Class 1 |
| H-B-221a, Type II .. | — | Type IV, Class 2 |
| H-B-221a, Type III . | — | Type IV, Class 3 |

6.7 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

*Rail*:

Brushes, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

*Motor*:

Brushes, not otherwise indexed.

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**MILITARY INTERESTS:**

**Army—Q**

**Navy—S**

**Air Force—MOA**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.